Luis Quesada

Senior Engineering Manager

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SUMMARY

26+ years coding, 21+ years managing systems, 11+ years in industry, and 7+ years managing people
Technical expertise in distributed systems and user-facing products, strategic vision, and can-do attitude
I drive fast, impactful solutions, tackling complex problems with wide consensus and innovation

EXPERIENCE

Google, 2014 – currently

YouTube Infrastructure, Software Engineering — Senior Engineering Manager (L7)

- Improved the performance of high-throughput pipelines by 1000x. **Award**: Performance Excellence
- Productionized and applied Large Language Models to increase the expressiveness of a product

Cloud Artificial Intelligence, Site Reliability Engineering — Senior Engineering Manager (L5→L7)

- Proposed to revamp capacity, monitoring, rollouts, data integrity, and frameworks across the entire developer organization, garnered top-down support and influenced the work of 50+ engineers to deploy the changes bottom-up, improving velocity and reliability on tens of products and >1K engineers. **Awards**: Cloud + Core + Google Tech Impact, Perfy, and 2x Tech Debt Buster
- Led the productionization of three large Cloud Artificial Intelligence products across reliability, scalability, security, and process requirements, and enabled their launch.

Datacenter Software, Site Reliability Engineering — Technical Leader/Engineering Manager (L5)

- Took over a project to revamp critical datacenter systems and landed it one year ahead of time.
- Troubleshot and addressed incidents with company-wide impact as part of on-call responsibilities.

Apps Storage, Site Reliability Engineering — Technical Leader (L4→L5)

- Improved the availability and efficiency of a widely-used internal storage system by implementing an A/B-testing solution to detect regressions and integrating it in the rollout automation with push-on-green and alerting capabilities.
- Responsible for resource provisioning and capacity management for the storage backends of external facing products with >1B users. Developed, tested, and deployed a forecasting and provisioning model that led to significant resource savings and increased reliability. **Award**: Feats of Engineering
- Responded to time-critical user-facing outages as part of deeply technical on-call responsibilities, and planned and performed high-risk high-complexity operations on live systems.

University of Granada, 2010 – 2014

Department of Computer Science and Artificial Intelligence — Research Fellow

- Developed an on-the-fly compiler of compilers that produces lexical analyzers and parsers with ambiguity support from a hierarchy of Java classes and resolves lexical and syntactic ambiguities by applying syntactic constraints, semantic constraints, and probabilities. Researched its application to multilingual Natural Language Processing and to language prototyping. Directed a Master Thesis on a language for definition of music pieces. **Award**: Best Thesis (as thesis director)
- Developed an unsupervised markerless 3-degree-of-freedom real-time motion tracking heuristic technique that required a single low-budget camera.

STUDIES **University of Granada**, 2004 – 2010

Master in Research, Soft Computing and Intelligent Systems. GPA 9.3/10

Bachelor of Science, Information Systems Engineering. GPA 8.7/10. **Award**: First of Class

Master of Science, Computer Science. GPA 9.2/10. **Awards**: First of Class, National Award, Best Thesis

Bachelor of Science, Computer Systems Engineering. GPA 8.7/10. **Award**: First of Class

SKILLS Systems: Architecture, performance, automation, monitoring, load balancing, capacity, and data integrity

Artificial Intelligence: Infrastructure and solutions, applied Generative Multi-modal Models

Computer Languages: Go, Java, Python, JavaScript, TypeScript, PHP, Bash shell script, and SQL

People: Organization design, career development, team health, coaching, presentation, and negotiation

Languages: Spanish (Native), English (C2), German (B2), Swiss German (B1), and Esperanto (B1)

PUBLICATIONS

Distributed Systems

- Published a tech talk and an article on capacity management
- Published a tech talk on Google's production environment
- Published a tech talk on the Paxos algorithm

Generative Artificial Intelligence

- Developed inpainting ComfyUI nodes
- Developed interactive UI ComfyUI nodes
- Developed prompt combination ComfyUI nodes

Computer Vision

- Developed a 3D motion tracking solution that works on a single camera

Language Processors

- Developed lexical analyzers, parsers, and a model specification library with ambiguity support
- Designed a domain-specific language for music prototyping
- Developed a Java bytecode similarity detector

Videogames

- Developed a rogue-like videogame in JavaScript with no frameworks
- Developed a web-based multiplayer videogame in PHP+MySQL, with 10K+ players in the '00s
- Developed a shooter videogame in C for the Game Boy Advance console
- Developed a dungeon videogame prototype in Java with custom physics and graphic engines
- Developed a physics engine and basic taxonomy in GameMaker
- Developed a dungeon videogame prototype in Go with custom physics and inventory engines

Team Management

- Co-authored an O'Reilly book chapter on managing team overload

Language Learning Resources

- Aggregated a dictionary across five languages

Literature

- Authored Spanish books for children on monsters, school anecdotes and science fiction

Music

- Produced one and two indie rock, one and two synthesizer, and one and two piano albums